

POWER BI PROJECT REPORT

INDIA RETAIL SALES DATA ANALYSIS AND BUSINESS INTELLIGENCE DASHBOARD

Submitted By

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Project Title

India Retail Sales Data Analysis Using Microsoft Power BI

Tools Used

- Microsoft Power BI Desktop
- Power Query
- DAX
- Microsoft Excel

Academic Year

2026–2027

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1. INTRODUCTION

Data has become an important resource for modern businesses. Organizations generate large amounts of information through sales transactions, customers, products, locations and business operations. However, raw data by itself does not provide useful information unless it is properly analyzed.

Data analytics helps organizations convert raw data into meaningful information and insights that can support better business decisions.

Microsoft Power BI is a Business Intelligence and data visualization tool developed by Microsoft. It allows users to connect different data sources, clean and transform data, create data models, perform calculations using DAX, and present information through interactive dashboards and reports.

This project focuses on analyzing an India Retail dataset using Microsoft Power BI. The analysis examines sales, profit, quantity, products, geographical performance, transaction values and sales trends. The dashboard is designed to provide a clear view of retail performance and identify areas where the business can improve revenue and profitability.

2. BACKGROUND OF THE PROJECT

The retail industry generates a large amount of transactional data every day. Understanding this data is important for identifying profitable products, high-performing locations, sales trends and areas requiring improvement.

The India Retail dataset used in this project contains retail transaction information including order priority, discount, unit price, freight expenses, freight mode, customer segment, product type, product sub-category, product container, state, city, region, order date, ship date, profit, quantity ordered and sales.

The project uses Power BI to transform this raw transactional data into an interactive Business Intelligence dashboard.

The dashboard helps analyze:

- Overall sales performance
- Profitability
- Product performance
- Regional performance
- Sales trends
- Transaction value
- Quantity sold
- Metro and non-metro sales classification
- Festive and non-festive sales
- City-level performance
- Relationship between price and sales volume

3. PROBLEM STATEMENT

The main problem addressed in this project is the difficulty of understanding retail performance from raw transactional data.

A large dataset containing sales, products, locations, quantities, prices and profits can be difficult to analyze manually.

The project therefore aims to develop an interactive Power BI dashboard that converts raw retail data into meaningful visual insights.

The analysis focuses on identifying:

- Which states and regions generate higher sales
- Which products contribute most to sales
- How sales change over time
- Which cities perform better or worse
- How price relates to sales volume
- How festive periods affect sales
- How different product categories contribute to revenue and profitability

4. OBJECTIVES OF THE PROJECT

The major objectives of the project are:

1. To analyze overall retail sales performance.
2. To identify high-performing and low-performing products.
3. To analyze sales performance across different states, cities and regions.
4. To examine sales trends over time.
5. To calculate important Key Performance Indicators (KPIs).
6. To calculate and analyze average transaction value.
7. To compare metro and non-metro sales using a defined sales-based classification.
8. To compare festive and non-festive sales periods.
9. To examine the relationship between unit price and quantity sold.
10. To identify business opportunities for improving revenue and profitability.

5. DATASET DESCRIPTION

The project uses an India Retail dataset containing **2,534 retail transaction records** and **18 columns**.

The dataset covers the period from **2010 to 2013**.

Dataset Columns

Column	Description
Order Priority	Priority assigned to the order
Discount Offered	Discount provided on the order
Unit Price	Price of one unit
Freight Expenses	Expenses associated with transportation
Freight Mode	Method of transporting the order
Segment	Customer/business segment
Product Type	Broad product category
Product Sub-Category	Specific product category
Product Container	Type of product packaging/container
State	State where the transaction occurred
City	City where the transaction occurred
Region	Geographical region
Country	Country associated with the transaction
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Profit	Profit generated from the transaction
QtyOrdered	Quantity of units ordered
Sales	Revenue generated from the transaction

Important Measures of the Dataset

The main business measures used in the analysis are:

- **Sales** – revenue generated from transactions
- **Profit** – profit generated from transactions
- **QtyOrdered** – number of units ordered
- **Unit Price** – price per unit
- **Discount Offered** – discount applied to transactions

6. DATA ANALYSIS LIFE CYCLE

The project follows a structured data analysis process.

1. Define the Problem

The first step is to understand the business problem and determine what information is required from the dataset.

2. Collect Data

The India Retail dataset was used as the primary source of transactional information.

3. Clean and Prepare Data

The dataset was checked for data quality issues, appropriate data types and consistency.

4. Explore the Data

The data was examined to understand sales, profit, products, locations and trends.

5. Analyze the Data

Measures and calculations were created to identify patterns and relationships.

6. Visualize the Data

Power BI visuals were used to present the analysis in an understandable form.

7. Generate Insights

The visualizations were interpreted to identify important business findings.

8. Business Decision Making

The findings can be used to support decisions related to products, locations, sales and profitability.

Overall Process

Raw Data → Cleaning → Analysis → Visualization → Insights → Business Decisions

7. DATA CLEANING AND TRANSFORMATION

Data cleaning was performed using **Power Query in Microsoft Power BI**.

The purpose of data cleaning was to make the dataset suitable for analysis and visualization.

Major Data Preparation Activities

7.1 Data Type Checking

The data types of columns were checked to ensure that:

- Dates were recognized as dates
- Sales and profit were recognized as numerical values
- Quantity was recognized as numerical data
- Text fields were recognized as text

7.2 Missing Value Checking

The dataset was checked for missing or blank values.

Missing values were reviewed before deciding whether they should be retained or handled.

7.3 Duplicate Checking

The dataset was checked for duplicate records to avoid incorrect calculations and double counting.

7.4 Date Preparation

Order Date was used as the main date field for time-based analysis.

A separate calendar table was created for year, month and quarter analysis.

7.5 Data Transformation

Power Query was used to prepare the data for analysis and ensure that the columns were suitable for Power BI calculations and visuals.

8. DATA MODELLING

Data modelling is the process of organizing data tables and relationships so that Power BI can analyze the data efficiently.

A calendar table was created for time-based analysis.

Calendar Table

The calendar table contains:

- Date
- Year
- Month Number
- Month
- Quarter
- Day

The calendar table was connected to the retail table using the Order Date.

Relationship

The relationship used was:

Calendar[Date] → retails[Order Date]

This relationship enables analysis by:

- Year
- Month
- Quarter
- Date

Importance of Data Modelling

A proper data model helps:

- Improve report performance
- Avoid incorrect calculations
- Support time-based analysis
- Make DAX calculations easier
- Provide consistent filtering across visuals

9. DAX MEASURES

DAX stands for **Data Analysis Expressions**. It is the formula language used in Power BI for creating calculations and measures.

The following measures were created for the project.

9.1 Total Sales

Total Sales =

SUM(retails[Sales])

This calculates the total revenue generated from all transactions.

9.2 Total Profit

Total Profit =

SUM(retails[Profit])

This calculates the total profit generated from the transactions.

9.3 Total Quantity

Total Quantity =

SUM(retails[QtyOrdered])

This calculates the total number of units ordered.

9.4 Total Transactions

A transaction ID was created using an index column because the original dataset did not contain a dedicated transaction ID.

Total Transactions =

COUNT(retails[Transaction ID])

This measure counts the number of transactions.

9.5 Average Transaction Value

Average Transaction Value =

DIVIDE(
[Total Sales],
[Total Transactions]
)

Average Transaction Value represents the average sales value generated per transaction.

10. DASHBOARD DEVELOPMENT

The Power BI dashboard was developed using different visualizations to answer specific business questions.

The major visuals used include:

- KPI Cards
- Bar Charts
- Line Charts
- Ribbon Charts
- Scatter Charts
- Tables
- Slicers

10.1 KPI Cards

The dashboard includes important KPIs such as:

- Total Sales
- Total Profit
- Total Quantity
- Total Transactions
- Average Transaction Value

These KPIs provide a quick overview of overall business performance.

10.2 Sales by State

A bar chart was used to analyze total sales across different states.

Purpose:

To identify states generating higher and lower sales.

10.3 Top 10 Product Sub-Categories by Sales

A bar chart was used to display the top 10 product sub-categories based on sales.

Purpose:

To identify products contributing significantly to revenue.

10.4 Sales Trend

A line chart was used to analyze sales trends over time.

Purpose:

To understand how sales changed across different periods.

10.5 Product Type Growth

A line chart was used to compare sales trends of different product types across years.

Purpose:

To identify product types showing growth or decline over time.

10.6 Metro vs Non-Metro Analysis

A sales-based classification was created to compare transactions based on sales value.

Metro Status =

IF(

retails[Sales] > 500,

"Metro",

"Non-Metro"

)

A Ribbon Chart was used to compare the relative sales contribution of Metro and Non-Metro classifications over time.

Note: This is a project-defined sales-based classification and does not represent an official geographical definition of metro cities.

10.7 Festive vs Non-Festive Analysis

A classification was created using October and November as the defined festive period.

Festive Period =

```
IF(  
    MONTH(retails[Order Date]) IN {10, 11},  
    "Festive",  
    "Non-Festive"  
)
```

This classification was used to compare sales during festive and non-festive periods.

10.8 City Performance

Cities were ranked based on total sales.

City Rank =

```
RANKX(  
    ALL(retails[City]),  
    CALCULATE([Total Sales]),  
    ,  
    DESC  
)
```

This allows identification of high-performing and lower-performing cities.

10.9 Price vs Sales Volume

A scatter chart was used to examine the relationship between:

- Unit Price

- Quantity Ordered
- Product Sub-Category
- Sales

Purpose:

To understand whether differences in product prices are associated with differences in sales volume.

11. ANALYSIS AND FINDINGS

The Power BI analysis provided several important findings from the retail dataset.

11.1 Overall Sales and Profit

The dataset generated approximately **₹2.515 million in total sales** and approximately **₹285,268 in total profit**.

The overall profit margin was approximately **11.34%**.

This provides a baseline for evaluating regional, product and segment performance.

11.2 Regional Performance

The analysis showed differences in sales and profitability across regions.

The **North region generated approximately ₹956.6K in sales**, while the **West region generated approximately ₹942.0K**.

Although sales were relatively high in these regions, profitability differed between regions.

The **South region showed the highest regional profit margin at approximately 14.4%**, while the West region had a lower margin of approximately 8.5%.

This shows that higher sales do not necessarily result in higher profitability.

11.3 Product Type Performance

The analysis showed that **Canned Foods generated approximately ₹1.064 million in sales**, making it a significant contributor to overall revenue.

Processed Meat generated approximately **₹117.3K in profit**, with a margin of approximately **21.2%**.

This indicates that product-level analysis is important because products can differ significantly in their contribution to profit.

11.4 Product Sub-Category Performance

The analysis showed differences in profitability among product sub-categories.

Quail Eggs generated approximately ₹70.9K in profit, while Jelly Fish recorded a profit margin of approximately 36.1%.

At the same time, some sub-categories such as **Tuna and Jams were loss-making.**

This demonstrates that high sales volume alone does not necessarily indicate strong profitability.

11.5 Product Container Performance

The analysis also showed differences based on product container.

Small Box generated approximately ₹181.2K in profit, while Wrap Bag recorded a profit margin of approximately 26.9%.

Some container categories, including **Large Box and Jumbo Box**, showed weak or negative profitability.

This may indicate differences in product mix, cost structure or pricing.

11.6 Year-wise Profit Performance

The analysis showed that yearly profit was approximately:

Year Profit

2010 ₹79.1K

2011 ₹79.1K

2012 ₹78.4K

2013 ₹48.7K

A significant decline in profit was observed in 2013 compared with earlier years.

At the same time, sales increased, indicating that higher sales did not automatically translate into higher profit.

This makes 2013 an important period for further investigation into pricing, discounts, freight expenses and product mix.

12. KEY BUSINESS INSIGHTS

Based on the analysis, the following key insights were identified.

Insight 1: High Revenue Does Not Always Mean High Profitability

Some regions and products generated high sales but relatively lower profit margins.

Therefore, management should consider both **revenue and profitability** when evaluating performance.

Insight 2: Product Mix Has a Major Impact

Some product sub-categories generated significantly higher profits, while others were loss-making.

Understanding the product mix can help businesses allocate resources toward more profitable products.

Insight 3: Regional Performance Differs

Sales and profit margins varied across regions.

This indicates that regional strategies may need to be adjusted based on both revenue and profitability.

Insight 4: Profit Declined in 2013

The decline in profit during 2013 despite strong sales indicates that the business should investigate the reasons behind the reduction in profitability.

Possible areas for investigation include:

- Discounts
- Freight expenses
- Pricing
- Product mix
- Low-margin products

Insight 5: Transaction-Level Analysis Supports Better Decisions

Analyzing individual transactions provides greater detail than looking only at total sales.

It helps identify patterns involving products, cities, quantities, prices and profitability.

13. RECOMMENDATIONS

Based on the findings from the Power BI analysis, the following recommendations can be considered.

13.1 Focus on Profitable Products

Management should identify products that consistently generate strong profit margins and evaluate opportunities to increase their sales contribution.

13.2 Review Loss-Making Products

Loss-making and low-margin product categories should be investigated.

Management can review:

- Pricing
- Procurement costs
- Discounts
- Freight expenses
- Product demand

before deciding whether to modify their strategy.

13.3 Improve Regional Strategies

Regions with lower profitability should be analyzed separately to understand the reasons behind their performance.

Different pricing, product and distribution strategies may be considered based on regional conditions.

13.4 Investigate the 2013 Profit Decline

The significant decline in profit during 2013 requires further analysis.

Management should examine whether increased discounts, higher freight costs, pricing changes or a shift toward low-margin products contributed to the decline.

13.5 Use Power BI for Continuous Monitoring

The dashboard can be updated regularly to monitor:

- Sales
- Profit

- Product performance
- Regional performance
- Quantity
- Transaction value

This can support faster and more data-driven decision making.

14. PROJECT LIMITATIONS

The project has some limitations based on the available dataset.

1. The dataset does not contain a dedicated original transaction ID.
2. The analysis is limited to the variables available in the dataset.
3. Actual customer demographic information is not available.
4. Population and income information was not available in the dataset, so it was not included in the analysis.
5. The Metro and Non-Metro classification used in the project is based on a defined sales threshold rather than an official geographical classification.
6. The Festive Period classification uses October and November as the defined festive period and does not represent exact festival dates.
7. The dataset covers the period from 2010 to 2013, so the findings are limited to this period.

15. FUTURE SCOPE

The project can be further enhanced in several ways.

15.1 Sales Forecasting

Machine learning or Power BI forecasting features can be used to estimate future sales.

15.2 Customer Analysis

Customer-level data can be added to analyze customer purchasing behavior and customer value.

15.3 Profit Forecasting

Future profitability can be estimated based on historical sales and cost patterns.

15.4 Real-Time Dashboard

The dashboard can be connected to live data sources to monitor business performance in real time.

15.5 Automated Data Refresh

Scheduled refresh can be implemented so that the dashboard automatically receives updated data.

15.6 Advanced Analytics

Advanced analytics techniques can be introduced to identify sales patterns, product relationships and potential business opportunities.

16. CONCLUSION

This project demonstrated how Microsoft Power BI can be used to transform raw retail transaction data into meaningful business insights.

The India Retail dataset was analyzed based on sales, profit, quantity, products, regions, cities, transactions and time periods. Power Query was used for data preparation, while DAX was used to create important business measures. Interactive Power BI visualizations were then developed to communicate the findings clearly.

The analysis showed that sales performance and profitability can vary significantly across products and regions. The project also identified the importance of analyzing product mix, regional performance and yearly profitability rather than focusing only on total sales.

Overall, the project demonstrates the value of Business Intelligence in helping organizations understand their data and make more informed, data-driven business decisions.

Power BI therefore provides an effective platform for converting retail data into interactive analysis, actionable insights and better business decision-making.